

KWAME NKRUMAH UNIVERSITY OF TECHNOLOGY



DEPARTMENT OF CIVIL ENGINEERING

COMPUTER PROGRAMMING (CE 257)

REPORT ON:

**APPLICATION DEVELOPMENT FOR TRUSS
DEFLECTION USNG PYTHON**

**BY
GROUP 9**

GROUP MEMBER	INDEX NUMBER
Elvis Aseye Fasemkye	3948824
Blessed Isaac Andoh	3947724
Ishmael Oduro	3950324

MAY 2026

ACKNOWLEDGEMENT

A special appreciation is extended to the Almighty God for His guidance, protection, and wisdom throughout the execution and completion of this project.

Sincere gratitude is also expressed to the lecturers of the Department of Civil Engineering, Dr. Jack Banahene Osei and Dr. Augustus Ababio, for their continuous guidance, academic support, and enlightenment throughout the semester. Their direction and instruction provided a strong foundation for the successful completion of this project.

Special thanks are further extended to senior coursemates for their willingness to assist, explain key concepts, and provide valuable support during the development of the project. Their contributions significantly enhanced the understanding and execution of the work.

ABSTRACT

This project focused on the development of a Python-based application for the analysis and visualization of statically determinate truss structures using the Method of Joints. The application was developed to simplify the process of truss analysis, reduce errors associated with manual calculations, and improve the speed and accuracy of structural computations.

The system was implemented using the Python programming language together with libraries such as Tkinter for the graphical user interface, NumPy for numerical computations, Matplotlib for visualization, and the Math library for geometric calculations. The application allows users to create truss geometries, define joints and members, apply loads, assign support conditions, and perform structural analysis through an interactive interface.

The Method of Joints was used to compute internal member forces and support reactions based on the principles of static equilibrium. The application also implemented the unit load method to determine truss deflections and display the deformed shape of the structure graphically.

To verify the accuracy of the developed system, several statically determinate truss configurations were tested and compared with manual calculations. The results obtained from the application showed complete agreement with the manual solutions, confirming the correctness and reliability of the program.

The developed application successfully achieved its intended objectives by providing both numerical and graphical outputs for truss analysis. The project also enhanced understanding of structural analysis, object-oriented programming, graphical user interface design, and computational problem-solving techniques.

TABLE OF CONTENT

ACKNOWLEDGEMENT	2
ABSTRACT	3
CHAPTER 1: INTRODUCTION	8
1.1 Background of Study	8
1.2 Problem Statement	8
1.3 Aim of Project	9
1.4 Objectives	9
1.5 Scope of Project	9
CHAPTER 2: LITERATURE REVIEW	10
2.1 Truss Structures	10
2.2 Statically Determinate Trusses	10
2.3 Method of Joints	11
CHAPTER 3: METHODOLOGY	12
3.1 Introduction	11
3.2 Programming Approach	12
3.3 Tools and Libraries Used	12
3.4 System Design	12
3.5 Program Flow	13
3.6 Mathematical Formulation	14
3.7 Software Implementation	14
3.8 Deflection Analysis (Unit Load Method)	14
CHAPTER 4: IMPLEMENTATION AND RESULTS	16
4.1 Introduction	16
4.2 System Implementation	16
4.3 Graphical User Interface (GUI)	16

4.4 Truss Modeling	17
4.5 Method of Joints Analysis	18
4.6 Software Implementation	19
4.7 Deflection Analysis (Unit Load Method)	19
4.8 Testing and Verification	20
4.8.1 Truss Configuration	21
4.8.2 Manual Verification Using Method of Joints	21
4.8.3 Comparison of Manual and Application Results	22
4.8.4 Deformed Shape Visualization	23
4.9 Results Obtained	23
CHAPTER 5: PROJECT MANAGEMENT AND IMPLEMENTATION PLAN	24
5.1 Project Implementation Plan	24
5.2 Work Distribution	25
CHAPTER 6: CONCLUSION AND RECOMMENDATION	26
6.1 Conclusion	26
6.2 Challenges Encountered	26
6.3 Recommendations	26
BIBLIOGRAPHY	27
APPENDICES	28

LIST OF TABLES

Table No.	Title	Page
Table 4.1	Joint Coordinates, Support Conditions and Applied Loads	21
Table 4.2	Member Properties	21
Table 4.3	Comparison of Member Forces	22
Table 4.4	Comparison of Support Reactions	23
Table 5.1	(No explicit title given – recommend “Project Implementation Plan (8- Week Schedule)”)	24

LIST OF FIGURES AND PLATES

Figure/Plate No.		Page
Figure 3.1	Flow Chart of Project	13
Plate 4.1	Main interface of the truss analysis application	17
Plate 4.2	Truss model with applied loads and support conditions	18
Plate 4.3	Solved truss showing member forces and support reactions	19
Plate 4.4	Deformed shape of the truss structure	20
Appendix A	Minutes of Meeting 1	28
Appendix B	Minutes of Meeting 2	28

CHAPTER 1: INTRODUCTION

1.1 Background of Study

Trusses are important structural systems commonly used in civil engineering for bridges, roofs, towers, and other load-bearing structures. They are preferred because they are strong, lightweight, and efficient in distributing loads through connected members.

The analysis of truss structures involves determining the forces in each member as well as the reactions at the supports when loads are applied. For statically determinate trusses, these forces can be found using basic equilibrium principles. One of the main methods used is the Method of Joints, where each joint in the structure is analyzed individually using the conditions of equilibrium.

In this method, the sum of forces in the horizontal and vertical directions at each joint must be zero.

$$\sum F_x = 0 \quad (1.1)$$

$$\sum F_y = 0 \quad (1.2)$$

While this method is straightforward, manual calculations can become difficult and time-consuming when dealing with larger truss systems. It also increases the chance of errors, especially when multiple joints and members are involved.

With the use of computer programming, these calculations can be automated to improve speed and accuracy. Python provides useful tools for computation and visualization, making it suitable for developing engineering applications.

This project therefore focuses on developing a Python-based application that applies the Method of Joints to analyze truss structures and display their deflected shapes. The application allows users to create truss models, apply loads, define supports, and view both numerical results and graphical representations.

1.2 Problem statement

The analysis of statically determinate truss structures is an essential part of structural engineering, as it helps determine internal member forces and support reactions under applied loads. Although manual methods such as the Method of Joints are well established and reliable, they become increasingly tedious and time-consuming when applied to complex truss systems with multiple joints and members.

In addition, manual computations are prone to human error, especially when resolving forces into components and solving simultaneous equilibrium equations at each joint. This can lead to inaccurate results and inefficient analysis processes.

There is therefore a need for a computational tool that can simplify the process of truss analysis by automating calculations, reducing errors, and improving efficiency. Such a tool should also provide a visual representation of the truss structure and its deformed shape to enhance understanding of structural behaviour.

This project addresses this need by developing a Python-based application that implements the Method of Joints to analyse truss structures, compute member forces and support reactions, and visually display the deformed shape of the truss system.

1.3 Aim of Project

The aim of this project is to develop a Python-based application for the analysis of statically determinate truss structures using the Method of Joints and to provide a graphical visualization of the truss deformation under applied loads.

1.4 Objectives

The main objectives of the project are:

- To develop an interactive interface for creating truss structures.
- To enable users to define joints, members, loads, and support conditions.
- To apply the Method of Joints for determining internal member forces.
- To compute support reactions for statically determinate trusses.
- To implement a grid system for accurate truss drawing and positioning.
- To visualize both the original and deformed shape of the truss structure.

1.5 Scope of Project

This project is limited to the analysis of two-dimensional statically determinate truss structures. The analysis is carried out using only the Method of Joints, based on the principles of static equilibrium.

The application allows users to create truss models, assign loads, and define support conditions such as pin and roller supports. It also computes internal member forces and support reactions and presents the results graphically.

The project does not cover statically indeterminate trusses, three-dimensional truss systems, or advanced methods such as the stiffness matrix method. The focus is strictly on visualization and analysis of simple truss systems using fundamental equilibrium equations.

CHAPTER 2 : LITERATURE REVIEW

2.1 Truss Structures

A truss is a structural system composed of straight members connected at joints to form a stable framework. These members are usually arranged in triangular shapes to ensure stability and efficient load transfer.

In engineering practice, trusses are widely used in bridges, roofs, transmission towers, and similar structures where long spans and lightweight construction are required. The behavior of a truss is based on the assumption that loads are applied only at the joints, causing the members to experience either tensile or compressive axial forces.

2.2 Statically Determinate Trusses

A truss is said to be statically determinate when all its internal member forces and support reactions can be determined using only the equations of static equilibrium. This means that the number of unknowns does not exceed the number of available equilibrium equations.

For a two-dimensional system, the equilibrium conditions are given as:

$$\sum M=0 \quad (2.1)$$

Eqns (1.1) and (1.2)

These equations ensure that the structure is in equilibrium under applied loads.

2.3 Method of Joints

The Method of Joints is a classical approach used in truss analysis to determine internal member forces. In this method, each joint is analyzed separately as a free body, and the forces acting on it are resolved into horizontal and vertical components.

For each joint, the equilibrium conditions must satisfy Eqns (1.1) and (1.2).

By solving these equations sequentially from joint to joint, the forces in all members of a statically determinate truss can be determined. Members in tension are assumed to pull away from the joint, while members in compression push toward the joint.

CHAPTER 3: METHODOLOGY

3.1 Introduction

This chapter explains the approach used in developing the truss analysis application. It describes the programming methodology, system design, program structure, and the implementation of the Method of Joints used for solving statically determinate trusses. It also outlines how the user interacts with the system to build and analyze truss structures.

3.2 Programming Approach

The application is developed using the Python programming language due to its simplicity and strong support for numerical computation and graphical visualization.

The system follows an object-oriented programming approach where the main structural components of a truss are represented as classes. This allows the model to closely reflect real engineering elements such as joints and members.

3.3 Tools and Libraries Used

The following tools and libraries were used in the development of the application:

- Tkinter: For building the graphical user interface (GUI), allowing user interaction such as adding joints, members, loads, and supports.
- Matplotlib: For graphical visualization of the truss structure and its deformed shape.
- NumPy: For numerical operations and mathematical computations.
- Math Library: For trigonometric and geometric calculations such as lengths and angles of members.

3.4 System Design

The system is designed around three main components:

(i) Joint Module

Each joint represents a connection point in the truss. It stores:

- ✓ Coordinates (x, y)
- ✓ Applied loads (P_x, P_y)
- ✓ Support conditions (pin or roller)

(ii) Member Module

Each member connects two joints and stores:

- ✓ Start and end joints
- ✓ Material properties (Young modulus and Area)
- ✓ Internal force (tension or compression)
- ✓ Geometric properties such as length and direction

(iii) Solver Module

This module performs structural analysis using the Method of Joints. It calculates:

- ✓ Internal member forces
- ✓ Support reactions
- ✓ Equilibrium at each joint

3.5 Program Flow

By combining a user-friendly interface with the precision of the **Method of Joints**, the application allows users to build structures from scratch, apply real-world forces, and observe how those structures react. The flow chart in Figure 3.1 provides a simple roadmap of this process, showing how the program moves from a user's initial drawing on a grid to the final calculation of member forces and the resulting deformed shape

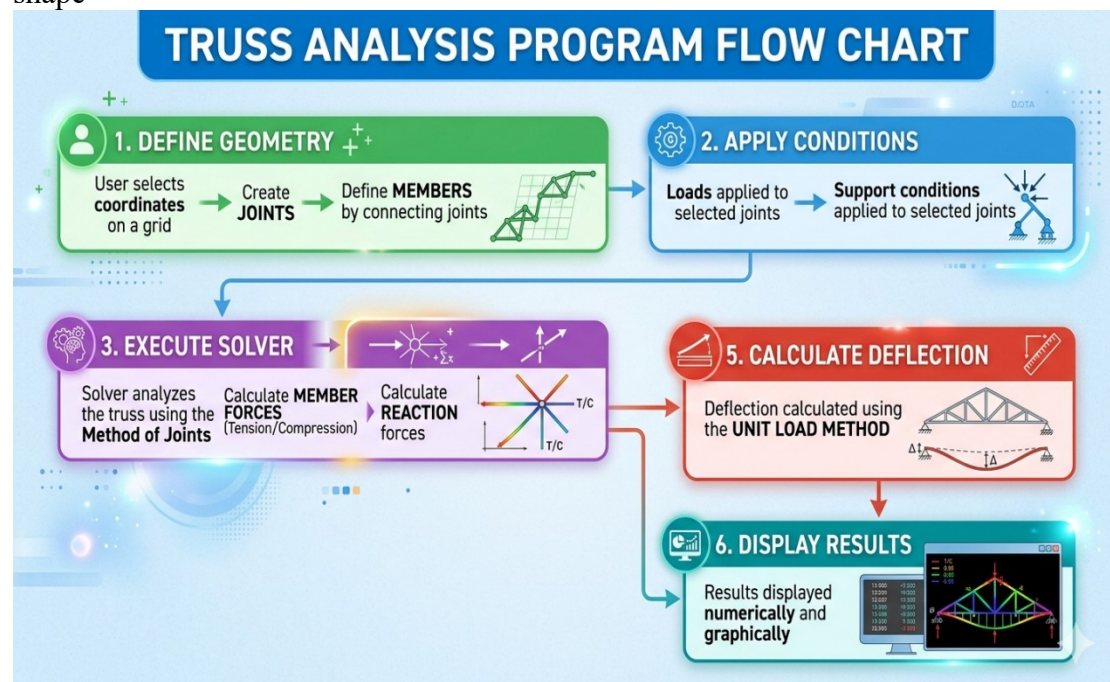


Figure 3.1. Flow Chart of Project

3.6 Mathematical Formulation

The analysis of each joint is based on static equilibrium principles. For a joint to be in equilibrium, the sum of forces in both horizontal and vertical directions must be zero, as shown in Eqns (1.1) and (1.2)

Each member force is resolved into components using direction cosines:

$$\cos\theta=\Delta x/L \quad (3.1)$$

$$\sin\theta=\Delta y/L \quad (3.2)$$

where:

- ✓ Δx and Δy are the differences in coordinates between joints
- ✓ (L) is the length of the member

3.7 Software Implementation

The application is implemented using an object-oriented structure consisting of the following main classes:

- ✓ Joint Class: Stores joint coordinates, loads, and support conditions.
- ✓ Member Class: Defines connections between joints and calculates geometric properties.
- ✓ MethodOfJointsSolver Class: Performs iterative equilibrium analysis to determine unknown forces.
- ✓ TrussApp Class: Handles the graphical user interface, user interactions, and visualization.

The solver operates iteratively by analyzing each joint repeatedly until all unknown member forces and reactions are determined.

3.8 Deflection Analysis (Unit Load Method)

Deflection of the truss is calculated using the unit load (virtual work) method. A unit load is applied at a joint in the direction of interest, and the resulting internal forces are used to compute displacement.

The general expression used is:

$$\Delta=\sum(F_v.\delta) / EA \quad (3.3)$$

where:

- ✓ (F_v) = virtual member force
- ✓ δ = real member deformation
- ✓ E = Young's Modulus
- ✓ A = Cross Sectional Area

This allows the application to compute horizontal and vertical deflections at each joint.

CHAPTER 4: IMPLEMENTATION AND RESULTS

4.1 Introduction

This chapter presents the implementation and results of the truss analysis application developed using Python. The chapter explains the system implementation process, graphical user interface, truss modeling procedures, force analysis, and deflection visualization produced by the application.

4.2 System Implementation

The application was implemented using Python programming language together with libraries such as Tkinter, NumPy, Matplotlib, and Math. The program was developed using an object-oriented structure to improve organization and functionality.

The system was divided into different sections including:

- ✓ Joint creation
- ✓ Member connectivity
- ✓ Load application
- ✓ Support definition
- ✓ Truss analysis
- ✓ Deflection visualization

4.3 Graphical User Interface (GUI)

The graphical user interface was developed using Tkinter. The interface allows users to interact with the application through buttons and a drawing area.

The GUI provides the following features:

- ✓ Grid system for drawing truss geometry
- ✓ Joint creation and positioning
- ✓ Member connection between joints
- ✓ Load application at joints
- ✓ Pin and roller support definition
- ✓ Display of member forces and reactions
- ✓ Visualization of original and deformed truss shapes

The interface also provides a results panel where calculated forces and reactions are displayed after analysis.

The interface is shown in plate 4.1.

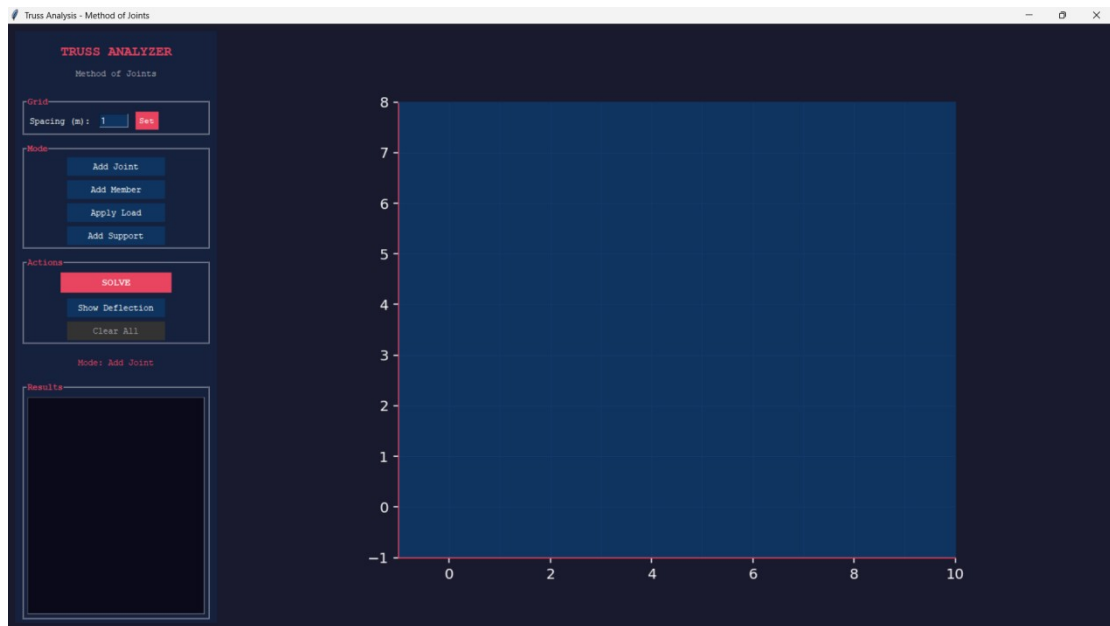


Plate 4.1: Main interface of the truss analysis application

4.4 Truss Modeling

The application allows users to create truss systems by placing joints on the grid and connecting them using members.

Each joint stores:

- ✓ Joint coordinates
- ✓ Applied loads
- ✓ Support conditions

Each member stores:

- ✓ Start joint
- ✓ End joint
- ✓ Young's Modulus (E)
- ✓ Cross-sectional Area (A)
- ✓ Member force

The truss geometry is visualized graphically to help users easily define the structure before analysis.

Plate 4.2 shows a truss drawn on the grid with joints, members, loads and supports added.

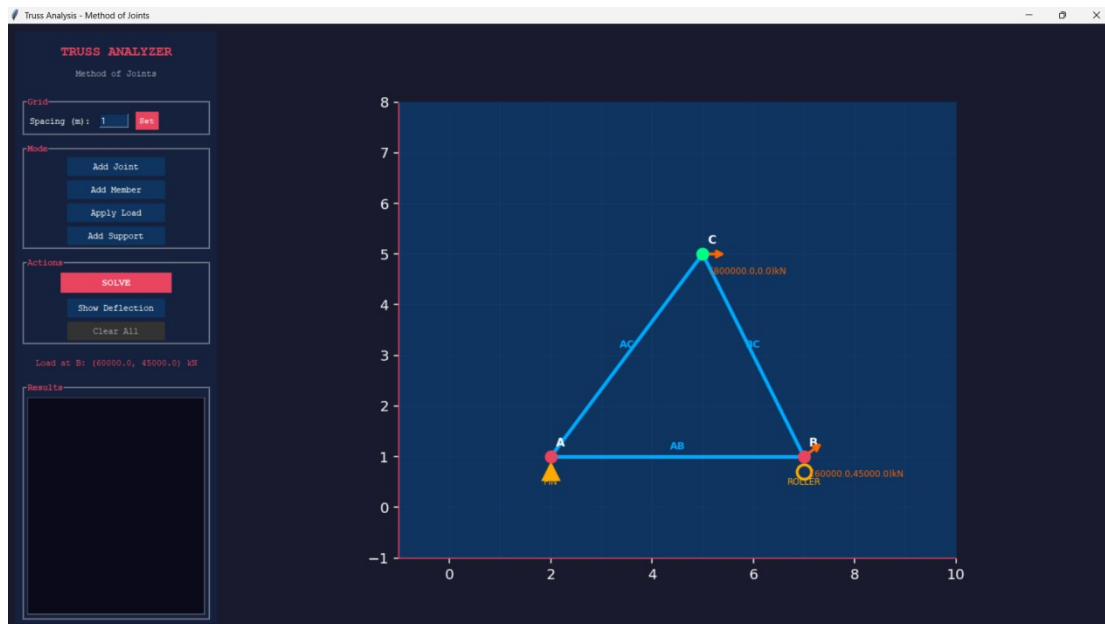


Plate 4.2: Truss model with applied loads and support conditions

4.5 Method of Joints Analysis

The Method of Joints was used to analyze the statically determinate truss system. The method applies equilibrium equations at each joint to determine unknown member forces and support reactions.

The equilibrium equations used are shown in Eqns (1.1) and (1.2)

The program resolves member forces into horizontal and vertical components using trigonometric relationships.

For a member inclined at an angle θ :

$$\cos \theta = \text{Horizontal Distance} / \text{Member Length}$$

$$\sin \theta = \text{Vertical Distance} / \text{Member Length}$$

The solver repeatedly analyzes each joint until all unknown member forces and support reactions are determined.

Plate 4.3 shows the results panel after clicking Solve — members coloured red/green showing compression/tension, with force values displayed

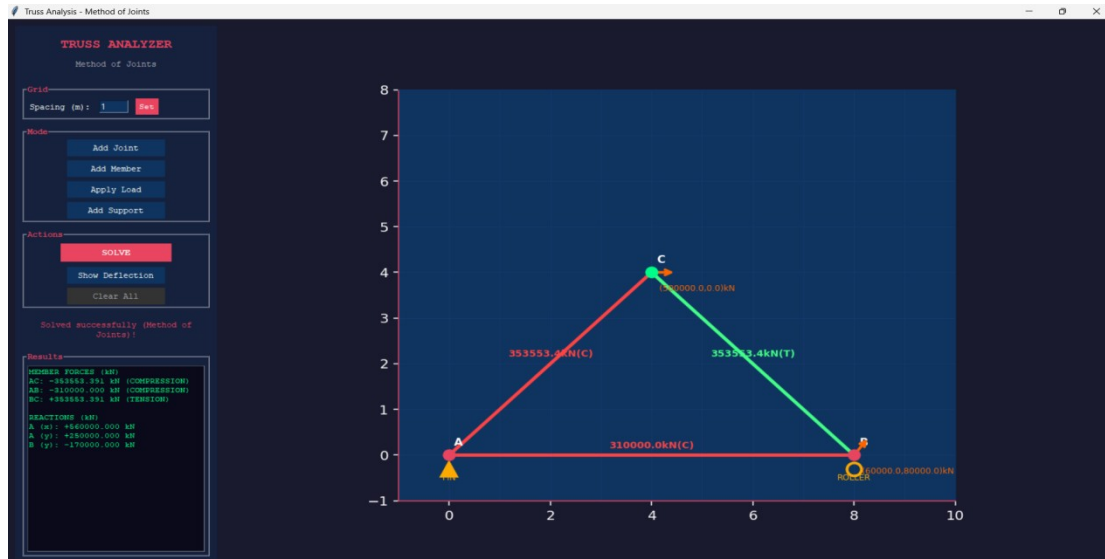


Plate 4.3: Solved truss showing member forces and support reactions

4.6 Software Implementation

The application was implemented using an object-oriented programming structure consisting of the following main classes:

- ✓ Joint Class: Stores joint coordinates, loads, and support conditions.
- ✓ Member Class: Defines connections between joints and calculates geometric properties.
- ✓ MethodOfJointsSolver Class: Performs iterative equilibrium analysis to determine unknown forces.
- ✓ TrussApp Class: Handles the graphical user interface, user interactions, and visualization.

The solver operates iteratively by analyzing each joint repeatedly until all unknown member forces and reactions are solved.

4.7 Deflection Analysis (Unit Load Method)

Deflection of the truss was calculated using the unit load (virtual work) method. A unit load is applied at a joint in the direction of interest, and the resulting internal forces are used to compute displacement.

The deformation equation used is:

$$\Delta = \sum (FfL / AE) \quad (4.1)$$

Where:

- ✓ Δ = Deflection
- ✓ F = Real member force
- ✓ f = Virtual member force due to unit load
- ✓ L = Length of member
- ✓ A = Cross-sectional area
- ✓ E = Young's Modulus

The computed deflections are displayed graphically by plotting both the original and deformed truss structures.

Plate 4.4 shows the deformed shape window showing the original truss (dashed) and the deformed truss (solid red)

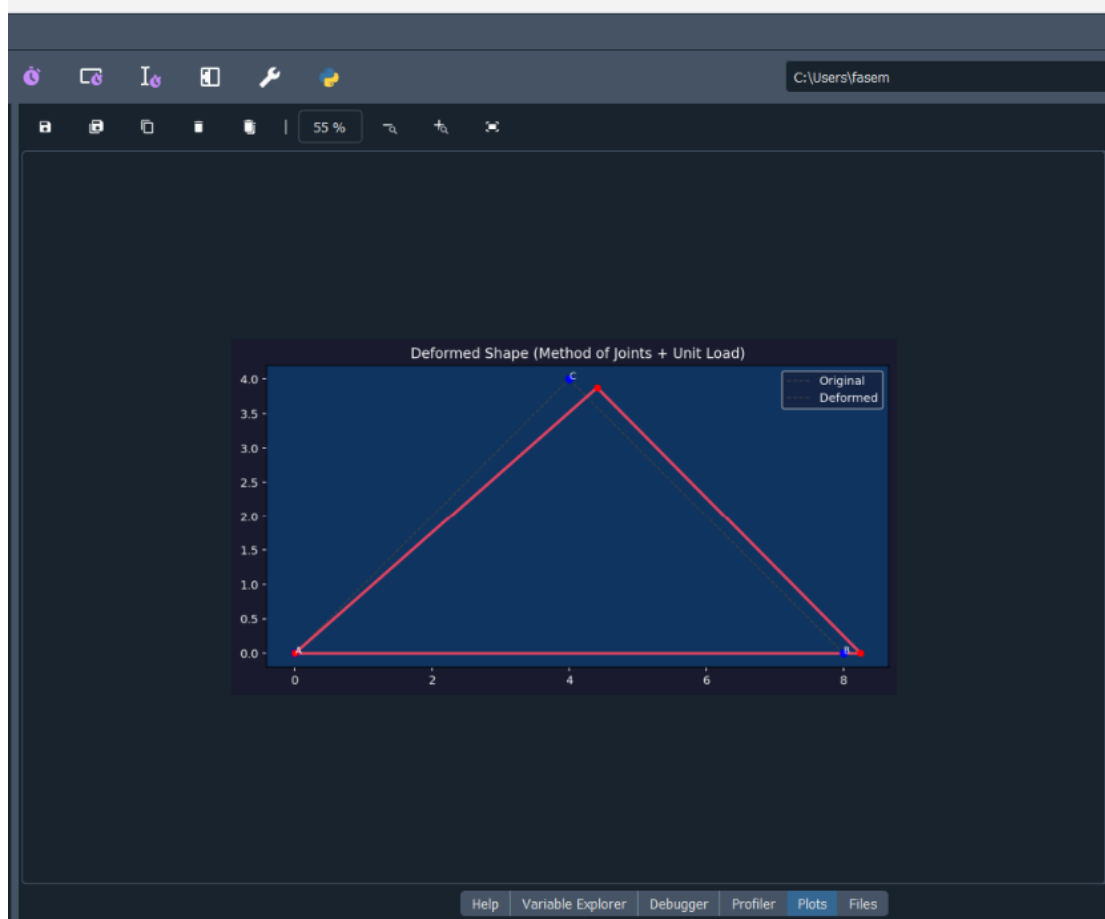


Plate 4.4: Deformed shape of the truss structure

4.8 Testing and Verification

The application was tested using different statically determinate truss configurations to verify the correctness of the analysis results.

The testing process involved:

- ✓ Applying different loads
- ✓ Changing support conditions
- ✓ Verifying member force directions
- ✓ Checking equilibrium at joints
- ✓ Comparing deformation patterns

Errors identified during testing were corrected to improve the performance and reliability of the application.

To verify the accuracy of the developed application, a simple statically determinate truss was analysed manually and the results compared with those obtained from the application. The truss configuration used for testing is described in Tables 4.1 and 4.2.

4.8.1 Truss Configuration

The test truss consists of three joints and three members. Joint A is located at coordinates (0, 0) and is assigned a pin support. Joint B is located at coordinates (4, 0) and is assigned a roller support. Joint C is located at coordinates (2, 2) and has no support condition. A vertical load of 10 kN was applied at joint C. The member properties are presented in Table 4.2.

Table 4.1: Joint Coordinates, Support Conditions and Applied Loads

Joint	X-Coordinate (m)	Y-Coordinate (m)	Support Condition	Applied Load (kN)
A	0.0	0.0	Pin	None
B	4.0	0.0	Roller	None
C	2.0	2.0	None	$P_y = 10 \text{ kN}$

Table 4.2: Member Properties

Member	Start Joint	End Joint	Length (m)	E (Pa)	A (m ²)
AC	A	C	2.828	2.0×10^{11}	0.01
BC	B	C	2.828	2.0×10^{11}	0.01
AB	A	B	4.000	2.0×10^{11}	0.01

4.8.2 Manual Verification Using Method of Joints

The method of joints was applied manually at Joint C to verify the application output.

Members AC and BC both make an angle of 45° with the horizontal, since the horizontal and vertical distances between joints are equal.

The inclination angle θ for both members:

$$\cos \theta = \sin \theta = 1/\sqrt{2} = 0.7071$$

Analysis at Joint C:

Applying the equilibrium conditions at Joint C, where a vertical load of 10 kN is applied:

Applying Eqns (1.1) and (1.2);

$$-F_{AC} \times \cos 45^\circ + F_{BC} \times \cos 45^\circ = 0 \quad \therefore$$

$$F_{AC} = F_{BC}$$

$$-F_{AC} \times \sin 45^\circ - F_{BC} \times \sin 45^\circ + 10 = 0 \quad -2 \times F_{AC} \times 0.7071 + 10 = 0 \quad F_{AC} = 10 / (2 \times 0.7071)$$

$$F_{AC} = F_{BC} = 7.071 \text{ kN (Tension)}$$

Analysis at Joint B:

At Joint B, the horizontal component of force in member BC and the force in member AB must be in equilibrium.

$$\Sigma F_x = 0:$$

$$F_{AB} - F_{BC} \times \cos 45^\circ = 0 \quad F_{AB} = 7.071 \times 0.7071 \quad F_{AB} = -5.000 \text{ kN (Compression)}$$

$$\Sigma F_y = 0:$$

$$R_{By} + F_{BC} \times \sin 45^\circ = 0 \quad R_{By} = -5.000 \text{ kN}$$

Analysis at Joint A:

$$\Sigma F_y = 0:$$

$$R_{Ay} + F_{AC} \times \sin 45^\circ = 0 \quad R_{Ay} = -5.000 \text{ kN}$$

$$\Sigma F_x = 0:$$

$$R_{Ax} + F_{AC} \times \cos 45^\circ - F_{AB} = 0 \quad R_{Ax} = 0.000 \text{ kN}$$

4.8.3 Comparison of Manual and Application Results

The results obtained manually were compared with those produced by the application. Table 4.3 presents the comparison of member forces and Table 4.4 presents the comparison of support reactions.

Table 4.3: Comparison of Member Forces

Member	Manual Result (kN)	Application Result (kN)	Condition	Agreement
AC	+7.071	+7.071	Tension	✓
BC	+7.071	+7.071	Tension	✓
AB	-5.000	-5.000	Compression	✓

Table 4.4: Comparison of Support Reactions

Support Direction		Manual Result (kN)	Application Result (kN)	Agreement
A	x	0.000	0.000	✓
A	y	-5.000	-5.000	✓
B	y	-5.000	-5.000	✓

The results from the manual calculation and those produced by the application are in complete agreement. This confirms that the Method of Joints solver implemented in the application is accurate and reliable.

4.8.4 Deformed Shape Visualization

Following the analysis, the unit load method was applied to compute the deflections at each joint. The application generated a graphical representation of both the original and deformed configurations of the truss, allowing for a clear visual comparison of the structural behaviour under the applied load.

4.9 Results Obtained

The developed application successfully performed the following functions:

- ✓ Creation of truss geometry using a grid system
- ✓ Application of loads and support conditions
- ✓ Determination of support reactions
- ✓ Calculation of member forces using the Method of Joints
- ✓ Visualization of the deformed shape of the truss

The application produced both numerical and graphical outputs, making the analysis process easier to understand and interpret.

CHAPTER 5: PROJECT MANAGEMENT AND IMPLEMENTATION PLAN

5.1 Project Implementation Plan

The project was completed over an eight-week period. Each week focused on specific tasks to ensure systematic development and successful completion of the application.

Table 5.1

Week	Activities / Tasks	Expected Outcome
Week 1	Understand the full project requirements and deliverables. Study the Method of Joints used for analyzing statically determinate trusses.	All group members clearly understand the objectives, analysis method, and responsibilities.
Week 2	Review basic truss equilibrium equations. Discuss project approach as a group and assign roles such as coding, documentation, and testing. Design the structure of the Python program. Define how joints, members, loads, and supports will be stored in the program. Create basic Python functions to store coordinates and member connectivity.	Initial program structure and data representation for the truss system completed.
Week 3	Implement a grid system that allows users to draw or define truss geometry. Allow the user to input joint coordinates and connect members. Use libraries such as NumPy or Matplotlib to help visualize the truss structure. Add functionality to apply loads at joints.	Program can generate and display a basic truss model from user input.
Week 4	Implement support conditions such as pin and roller supports. Develop the equilibrium equations required for each joint.	The truss model now includes loads and support reactions ready for analysis
Week 5	Implement the Method of Joints calculations. Solve simultaneous equilibrium equations to determine member forces. Test the program using simple truss examples to verify correctness.	Program successfully calculates forces in truss members.
Week 6	Implement visualization of the truss deformation. Compute and display the deformed shape of the truss structure. Plot both original and deformed configurations for comparison.	Program displays graphical results showing truss deformation.
Week 7	Test the software using multiple truss configurations. Fix coding errors and improve performance. Organize the project files and upload the Python scripts to a GitHub repository.	Stable and functional program available online with documentation
Week 8	Prepare the final group report explaining methodology, implementation steps, and results. Record individual videos demonstrating the coding process and explaining the program. Review all deliverables and prepare the final submission.	Complete project ready for submission including report, code, and recorded demonstrations.

The structured timeline was planned on meeting one(see Appendix A).

5.3 Work Distribution

Group Member Responsibility:

- ✓ Ishmael Oduro :Data structure development and equation formulation
- ✓ Elvis Fasemkye: Function development and coding
- ✓ Blessed Andoh: Research coordination and project management

Details are shown in the minutes of the meeting (see Appendix B).

CHAPTER 6: CONCLUSION AND RECOMMENDATION

6.1 Conclusion

The project successfully developed a Python-based application capable of analyzing statically determinate truss structures using the Method of Joints. The application allows users to create truss geometries, apply loads, define support conditions, compute member forces, and visualize the deformed shape of the truss structure.

The use of Python libraries such as Tkinter, NumPy, and Matplotlib helped provide both numerical and graphical outputs for the analysis process. The implementation of the Method of Joints enabled accurate determination of internal member forces and support reactions, while the unit load method was used to calculate truss deflections.

The project also improved understanding of structural analysis concepts, programming logic, graphical user interface development, and computational problem-solving techniques. Overall, the objectives of the project were successfully achieved.

6.2 Challenges Encountered

Several challenges were encountered during the development of the application. These included:

- ✓ Difficulty in writing code to solve for deflection
- ✓ Difficulty in implementing the iterative solving process for unknown member forces
- ✓ Errors related to member connectivity and force directions
- ✓ Challenges in plotting the deformed truss shape accurately
- ✓ Debugging and correcting logical errors in the program

Despite these challenges, continuous testing and group collaboration helped improve the performance and accuracy of the system.

6.3 Recommendations

The following recommendations are suggested for future improvements of the project:

- ✓ Inclusion of analysis for statically indeterminate trusses
- ✓ Addition of automatic error detection for unstable truss structures
- ✓ Improvement of the graphical user interface for better user experience
- ✓ Addition of member stress and strain calculations
- ✓ Development of a more advanced visualization system with animation features
- ✓ Implementation of file saving and loading functionality for truss models

BIBLIOGRAPHY

- Hibbeler, R. C. (2017). Structural Analysis: In SI Units (9th ed., Global ed.). Pearson.
- Githuku, C. R. (2024, November 4). *ECV 305 Theory of Structures III: Deflection - Energy Methods - Trusses*. CliffsNotes. Retrieved May 2026.
- Matplotlib Development Team. (2026). Matplotlib: 2D and 3D plotting (Version 3.x) [Computer software]. Retrieved from [https://matplotlib.org\[reference:3\]](https://matplotlib.org[reference:3])
- Emhayki. (2026). *Truss-Solver*. GitHub repository. Available at: <https://github.com/emhayki/Truss-Solver>. Accessed May 2026.
- Codemy.com. (2026). *How to Add a Matplotlib Graph to Tkinter Window in Python*. YouTube video. Available at: https://www.youtube.com/watch?v=fwk_WFuYmoA. Accessed May 2026

APPENDICES

APPLICATION DEVELOPMENT FOR DETERMINATE TRUSSES

Date: 12th March, 2026.

Venue: Lemez Hostel

Time: 6:00pm

Agenda

1. Distribution of roles
2. Understanding the concept of the project

Present

1. Elvis Aseye Fasemkye
2. Blessed Isaac Andoh
3. Ishmael Oduro

ITEM	PROCEEDINGS	ACTION BY
1.0	OPENING 1. Chairman called the meeting and welcomed all present after the opening prayer by 6:00am 2. All members signed the attendance list	Ishmael Oduro
2.0	ANALYSIS OF PROJECT QUESTION	All present
3.0	ALLOCATION OF TASK 1. Ishmael was assigned the task of breaking down the question an generating equations 2. Elvis was assigned the task of writing the codes 3. 3. Blessed was assigned the task of doing research on method of joints	Elvis Fasemkye
4.0	CLOSURE In the absence of further deliberations, the meeting was brought to an end	Ishmael Oduro

Appendix A

APPLICATION DEVELOPMENT FOR DETERMINATE TRUSSES(2nd Meeting)

Date: 28th March, 2026.

Venue: Lemez Hostel

Time: 7:00pm

Agenda

To plan the structure and design of the Truss deflection Application

Present

1. Elvis Aseye Fasemkye
2. Blessed Isaac Andoh
3. Ishmael Oduro

ITEM	PROCEEDINGS	ACTION BY
1.0	OPENING Meeting commenced with all three members present. Purpose of the meeting was explained.	All Members
2.0	PROGRAM STRUCTURE DESIGN Agreed on the format on which various variables, functions, classes, loads members and joints would be stored	Elvis Fasemkye
3.0	INITIAL FUNCTION Some basic functions were created to store joint coordinates, define members and input loads/ support conditions	Blessed <u>Andoh</u>
4.0	ALLOCATION OF TASK 1. Ishmael was assigned the data structure developer 2. Elvis was assigned the <u>functioner</u> developer 3. 3. Blessed was assigned the programmer coordinator	Ishmael Oduro
5.0	CLOSURE In the absence of further deliberations, the meeting was brought to an end	Blessed <u>Andoh</u>

Appendix B